Institute of Technology of Cambodia
Department of Applied Mathematics and Statistics

Internship Management System

Presented By I3_AMS_A-Group 1

Lecturer: Mr. KHEAN Vesal

Member

Name	ID
CHAN Serey	e20230409
HACH Lyheng	e20231175
HAY Pechmanea	e20230391
HORN Somborokisin	e20230300
HORT Kimmeng	e20231111

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1 Introduction

The Internship Management System (WEBIN) is a web-based application designed to simplify the internship process for students, companies, and universities. The system allows students to apply for internships, companies to manage opportunities, and supervisors to monitor student progress. It helps improve efficiency, communication, and record management.

2 Objective

The primary objectives of the WEBIN database project are:

- **Centralized platform** for internship management
- **Data integrity** via FK constraints, CHECK, triggers
- **Process automation** — notifications, slot mgmt, expiry
- **Query performance** with strategic indexing
- **Normalized schema** (3NF) for scalability
- **Role-based access** — 4 user roles with distinct permissions

3 Storyline

A student's journey through WEBIN

1️⃣ Sok registers as a student, creates his profile
2️⃣ Browses active internships at Companies
3️⃣ Applies with resume & transcript documents
4️⃣ Companies reviews, schedules online interview
5️⃣ Application accepted → slot auto-updated
6️⃣ Dr. Sokhey assigned as supervisor
7️⃣ Midterm & Final evaluations submitted with auto-scoring

4 Entity and Attribute

19 tables in the WEBIN database

- UserAccount — Email, password, role (student/supervisor/company/admin)
- Student — Name, DOB, gender, nationality, GPA, major, year
- Supervisor — Department, position, specialization, status
- Company — Industry, verified_status (Pending/Verified/Rejected)
- Admin — Name, phone, status
- InternPosition — Title, salary range, slots, deadline, status
- Application — Cover letter, status (Pending→Accepted/Rejected)
- Interview — Schedule, type (Online/Onsite), status

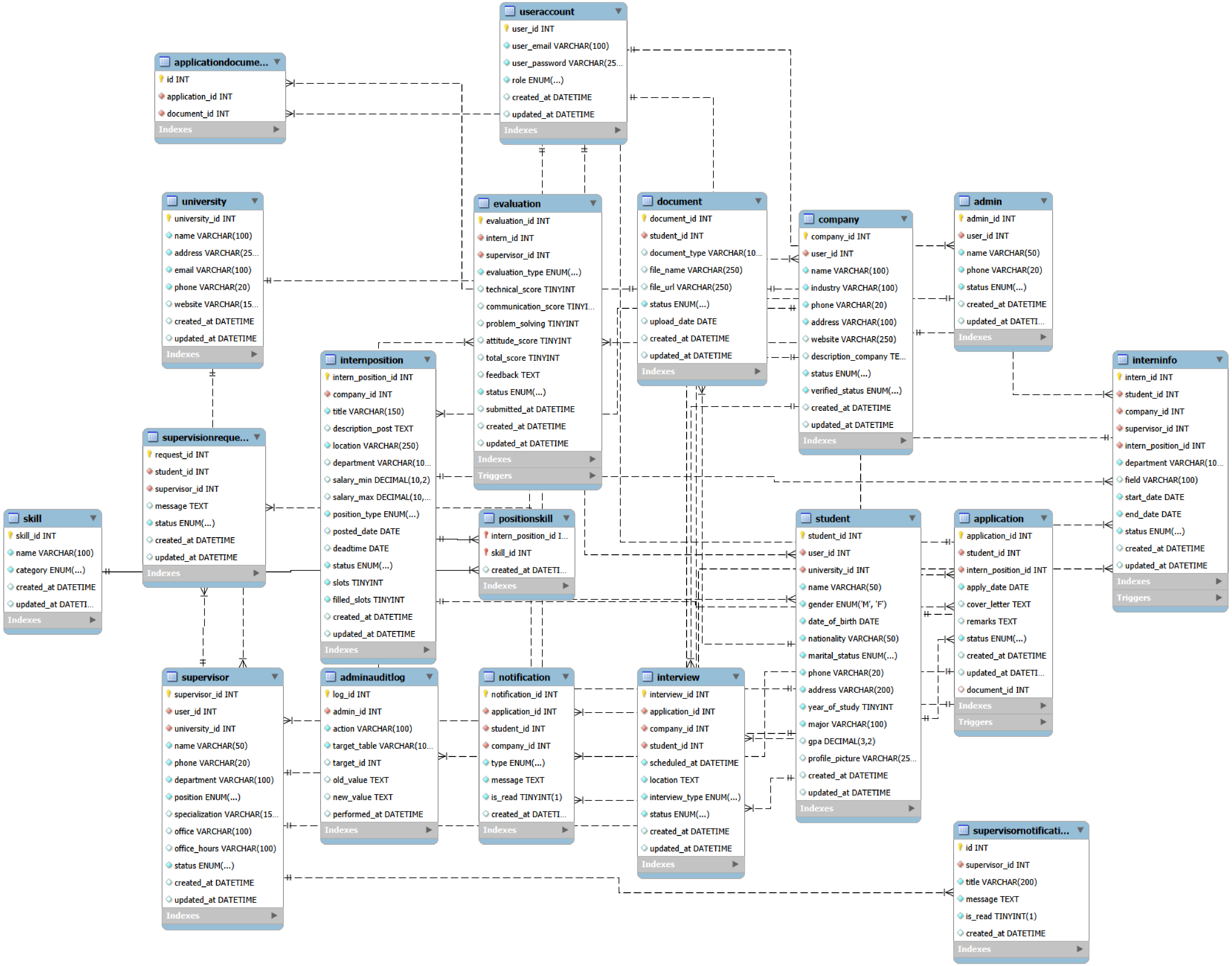
- InternInfo — Start/end dates, department, status
- Evaluation — Scores (technical, comm, problem-solving, attitude)
- Skill — Name + category (10 categories)
- Document — File name, type, verification status
- Notification — Type, message, read status
- SupervisionRequest — Student → Supervisor approval workflow
- PositionSkill — Junction: Position ↔️ Skill
- ApplicationDocument — Junction: Application ↔️ Document
- University — Name, address, contact info
- SupervisorNotification — Supervisor-specific alerts
- AdminAuditLog — Admin action history

5 Relationship

- UserAccount → 1:1 with Student / Supervisor / Company / Admin
- University → 1:M with Student, Supervisor
- Company → 1:M with InternPosition, Interview, InternInfo
- InternPosition → 1:M with Application; M:M with Skill via PositionSkill
- Student → 1:M with Application, Document, InternInfo
- Application → 1:1 with Interview; 1:M with Notification
- Supervisor → 1:M with InternInfo, Evaluation, SupervisionRequest
- InternInfo → 1:M with Evaluation (Midterm + Final)
- Admin → 1:M with AdminAuditLog

6 Entity Relationship Diagram

Entity Relationship Diagram (ERD) is a visual model used to design and represent a database structure. It shows the entities (tables), their attributes (columns), and the relationships between them. ERDs help developers understand how data is organized, connected, and stored before implementing the database.



useraccount
user_id INT
user_email VARCHAR(100)
user_password VARCHAR(25...
role ENUM(...)
created_at DATETIME
updated_at DATETIME
Indexes
applicationdocume...
id INT
application_id INT
document_id INT
Indexes
university
university_id INT
name VARCHAR(100)
address VARCHAR(25...
email VARCHAR(100)
phone VARCHAR(20)
website VARCHAR(15...
created_at DATETIME
updated_at DATETIME
Indexes
evaluation
evaluation_id INT
intern_id INT
supervisor_id INT
evaluation_type ENUM(...)
technical_score TINYINT
communication_score TINYI...
problem_solving TINYINT
attitude_score TINYINT
total_score TINYINT
feedback TEXT
status ENUM(...)
submitted_at DATETIME
created_at DATETIME
updated_at DATETIME
Indexes
Triggers
document
document_id INT
student_id INT
document_type VARCHAR(10...
file_name VARCHAR(250)
file_url VARCHAR(250)
status ENUM(...)
upload_date DATE
created_at DATETIME
updated_at DATETIME
Indexes
company
company_id INT
user_id INT
name VARCHAR(100)
industry VARCHAR(100)
phone VARCHAR(20)
address VARCHAR(100)
website VARCHAR(250)
description_company TE...
status ENUM(...)
verified_status ENUM(...)
created_at DATETIME
updated_at DATETIME
Indexes
admin
admin_id INT
user_id INT
name VARCHAR(50)
phone VARCHAR(20)
status ENUM(...)
created_at DATETIME
updated_at DATETI...
Indexes
interninfo
intern_id INT
student_id INT
company_id INT
supervisor_id INT
intern_position_id INT
department VARCHAR(10...
field VARCHAR(100)
start_date DATE
end_date DATE
status ENUM(...)
created_at DATETIME
updated_at DATETIME
Indexes
Triggers
internposition
intern_position_id INT
company_id INT
title VARCHAR(150)
description_post TEXT
location VARCHAR(250)
department VARCHAR(10...
salary_min DECIMAL(10,2)
salary_max DECIMAL(10,...
position_type ENUM(...)
posted_date DATE
deadtime DATE
status ENUM(...)
slots TINYINT
filled_slots TINYINT
created_at DATETIME
updated_at DATETIME
Indexes
supervisionreque...
request_id INT
student_id INT
supervisor_id INT
message TEXT
status ENUM(...)
created_at DATETIME
updated_at DATETIME
Indexes
skill
skill_id INT
name VARCHAR(100)
category ENUM(...)
created_at DATETIME
updated_at DATETI...
Indexes
positionskill
intern_position_id I...
skill_id INT
created_at DATETI...
Indexes
student
student_id INT
user_id INT
university_id INT
name VARCHAR(50)
gender ENUM('M', 'F')
date_of_birth DATE
nationality VARCHAR(50)
marital_status ENUM(...)
phone VARCHAR(20)
address VARCHAR(200)
year_of_study TINYINT
major VARCHAR(100)
gpa DECIMAL(3,2)
profile_picture VARCHAR(25...
created_at DATETIME
updated_at DATETIME
Indexes
application
application_id INT
student_id INT
intern_position_id INT
apply_date DATE
cover_letter TEXT
remarks TEXT
status ENUM(...)
created_at DATETIME
updated_at DATETI...
document_id INT
Indexes
Triggers
supervisor
supervisor_id INT
user_id INT
university_id INT
name VARCHAR(50)
phone VARCHAR(20)
department VARCHAR(100)
position ENUM(...)
specialization VARCHAR(15...
office VARCHAR(100)
office_hours VARCHAR(100)
status ENUM(...)
created_at DATETIME
updated_at DATETIME
Indexes
adminauditlog
log_id INT
admin_id INT
action VARCHAR(100)
target_table VARCHAR(10...
target_id INT
old_value TEXT
new_value TEXT
performed_at DATETIME
Indexes
notification
notification_id INT
application_id INT
student_id INT
company_id INT
type ENUM(...)
message TEXT
is_read TINYINT(1)
created_at DATETI...
Indexes
interview
interview_id INT
application_id INT
company_id INT
student_id INT
scheduled_at DATETIME
location TEXT
interview_type ENUM(...)
status ENUM(...)
created_at DATETIME
updated_at DATETIME
Indexes
supervisornotificati...
id INT
supervisor_id INT
title VARCHAR(200)
message TEXT
is_read TINYINT(1)
created_at DATETIME
Indexes

7 DDL - Table Creation

```
CREATE TABLE UserAccount (
    user_id      INT AUTO_INCREMENT NOT NULL,
    user_email   VARCHAR(100)      NOT NULL,
    user_password VARCHAR(255)      NOT NULL,
    role         ENUM('student','supervisor','company','admin') NOT NULL,
    created_at   DATETIME DEFAULT CURRENT_TIMESTAMP,
    updated_at   DATETIME DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP,
    CONSTRAINT pk_useraccount        PRIMARY KEY (user_id),
    CONSTRAINT uq_useraccount_email   UNIQUE     (user_email),
    CONSTRAINT ck_useraccount_email   CHECK (user_email LIKE '%@%.%'),
    CONSTRAINT ck_useraccount_password CHECK (CHAR_LENGTH(user_password) >= 8)
);
```

- 19 CREATE TABLE statements
- PK, FK, UNIQUE, CHECK constraints enforced
- ON DELETE RESTRICT + ON UPDATE CASCADE on all FKs

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DDL - Indexes, Triggers & Events

11 INDEXES

- InternPosition: status, company+status, deadline
- Application: status, student+status
- Notification: student+read, company+read
- InternInfo: status, supervisor+status
- Evaluation: status, supervisor+status
- Document: student+status
- AdminAuditLog: performed_at

7 TRIGGERS

- Validate application before insert
- Notify company on new application
- Notify student on status change
- Auto-manage filled_slots
- Auto-calculate evaluation total_score (insert + update)
- Auto-create Midterm & Final evaluations

1 SCHEDULED EVENT

- Daily: expire positions past deadline → status = "Expired"

9 DML – Data Seeding

INSERT / UPDATE examples

```sql
INSERT INTO university (university_id, name, address, email, phone, website) VALUES
(1, 'Royal university of Phnom Penh', 'Russian Federation Blvd, Phnom Penh', 'info@rupp.edu.kh', '023883640', 'https://www.rupp.edu.kh'),
(2, 'Institute of Technology of Cambodia', 'Russian Federation Blvd, Phnom Penh', 'info@itc.edu.kh', '023880370', 'https://www.itc.edu.kh'),
(3, 'National university of Management', 'St. 96, Phnom Penh', 'info@num.edu.kh', '023428120', 'https://www.num.edu.kh'),
(4, 'BELTEI International university', 'Phnom Penh, Cambodia', 'info@beltei.edu.kh', '023999967', 'https://www.beltei.edu.kh'),
(5, 'Cambodia Academy of Digital Technology', 'National Road 6A, Phnom Penh', 'info@cadt.edu.kh', '023722335', 'https://www.cadt.edu.kh');
```

```sql
INSERT INTO Skill (name, category) VALUES
        ("Python",        "Programming"),
        ("JavaScript",    "Programming"),
        ("Java",          "Programming"),
        ("C++",           "Programming"),
        ("React",         "Frontend"),
        ("Vue.js",        "Frontend"),
        ("HTML/CSS",      "Frontend"),
        ("Tailwind CSS",  "Frontend"),
        ("FastAPI",       "Backend"),
        ("Django",        "Backend"),
        ("Node.js",       "Backend"),
("Node.js",       "Backend"),
("REST API",      "Backend"),
("MySQL",         "DataBase"),
("PostgreSQL",    "DataBase"),
("MongoDB",       "DataBase"),
("SQLAlchemy",    "DataBase"),
("Docker",        "DevOps"),
("Git",           "DevOps"),
("AWS",           "DevOps"),
("Linux",         "DevOps"),
("Figma",         "Design"),
("Adobe XD",      "Design"),
("UI/UX Design",  "Design"),
("Power BI",      "Data"),
("Tableau",       "Data"),
("Excel",         "Data"),
("Pandas",        "Data"),
("Machine Learning", "AI_ML"),
("PyTorch",       "AI_ML"),
("TensorFlow",    "AI_ML"),
("scikit-learn",  "AI_ML"),
("Flutter",       "Mobile"),
("React Native",  "Mobile");
```

10 DQL — Data Queries

SELECT queries for reporting & dashboards

```
-- Active positions with company & remaining slots
SELECT ip.title, c.name AS company, ip.location,
       ip.salary_min, ip.salary_max,
       (ip.slots - ip.filled_slots) AS remaining
FROM InternPosition ip
JOIN Company c ON c.company_id = ip.company_id
WHERE ip.status = 'Active' ORDER BY ip.posted_date DESC;
```

```
-- Company application summary
SELECT ip.title,
       COUNT(*) AS total,
       SUM(CASE WHEN a.status='Pending' THEN 1 ELSE 0 END) AS pending,
       SUM(CASE WHEN a.status='Accepted' THEN 1 ELSE 0 END) AS accepted
FROM InternPosition ip
LEFT JOIN Application a ON a.intern_position_id = ip.intern_position_id
WHERE ip.company_id = 1 GROUP BY ip.title;
```

```
-- Student application history
SELECT a.application_id, ip.title, c.name,
       a.apply_date, a.status
FROM Application a
JOIN InternPosition ip ON ip.intern_position_id = a.intern_position_id
JOIN Company c ON c.company_id = ip.company_id
WHERE a.student_id = 1 ORDER BY a.apply_date DESC;
```

11 Tech Stack

- Database: MySQL
- Backend: Python FastAPI + SQLAlchemy 2.0
- Auth: JWT + bcrypt
- Frontend: React 19 / Vite 7 / Tailwind CSS v4
- API: RESTful JSON
- ERD Tool: MySQL Workbench

3-Tier Architecture

- Frontend (React) <-> API (FastAPI) <-> Database (MySQL)
- CORS configured for Vite dev server
- Connection pooling: pool_size=10, max_overflow=20

12 Conclusion

The Internship Management System streamlines the internship process by connecting students, companies, supervisors, and administrators in a single platform. It improves efficiency, data accuracy, and communication while providing a structured and reliable way to manage internship activities.

THANK YOU